

Algebraic Structures II Final

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Note $K := \mathbb{Q}[a_1, \dots, a_k]$ and $R := \mathbb{Z}[a_1, \dots, a_k]$ where $a_1, \dots, a_k$ are roots to F .

Problem 1. Explain why K is a field, and a finite extension of $\mathbb{Q}$. Explain why R is a finitely generated $\mathbb{Z}$ -module (ie a finitely generated Abelian group), and isomorphic as Abelian group to $\mathbb{Z}^d$ where d is the dim of K as a $\mathbb{Q}$ -vector space. Explain why K is the fraction field of R .

Proof. For $i = [1, \dots, k]$, define $K_i = \mathbb{Q}[a_1, \dots, a_i]$ and define $R_i := \mathbb{Z}[a_1, \dots, a_i]$.

Base Case: Since a_1 is the solution to F , a_1 is algebraic over $\mathbb{Q}$. Thus $\exists f_1(x) \in \mathbb{Q}[x]$, the minimal polynomial of a_1 . $f_1(x) = \sum_{k=0}^{\deg f_1} \frac{p_k}{q_k} x^k$. WLOG $\forall q_k = 1$, because $f_1(x) := \text{lcm}(q_1, \dots, q_{\deg f_1}) f_1(x)$. Thus $f_1(x) \in \mathbb{Z}[x]$, and since it is irreducible in $\mathbb{Q}[x]$, it is irreducible in $\mathbb{Z}[x]$. Thus, $K_1 = \mathbb{Q}[x]/(f_1)$ and $R_1 = \mathbb{Z}[x]/(f_1)$. Since a_1 is algebraic, K_1 is a finite extension over $\mathbb{Q}$ and $[K_1 : \mathbb{Q}] = \deg f_1$. Moreover $\{1, a_1, \dots, a_1^{\deg f_1}\}$ was a basis for R_1 moreover since f_1 was irreducible in $\mathbb{Z}$, that means, $\text{rank}_{\mathbb{Z}}(\mathbb{Z}[a_1]) = \deg f_1$. Moreover since $\mathbb{Z} \subset \mathbb{Q} \implies R_1 \subset Q_1$. Furthermore, $\forall k \in \mathbb{Q}[a_1]$, $k = \frac{p(a_1)}{q(a_1)}$ for $p(x), q(x) \in \mathbb{Z}[x] \ni q(a_1) \neq 0$ since $\mathbb{Q}$ is the fraction field of $\mathbb{Z}$, thus $k \in \text{Frac}(\mathbb{Z}[a_1])$. Thus $Q_1 \subseteq \text{Frac}(R_1)$. $\forall k \in \text{Frac}(R_1), k = \frac{r}{s}$ for $r, s \in R_1 \ni s \neq 0, k \in K_1$ by definition. Thus $\text{Frac}(R_1) \subseteq K_1 \implies R_1 = \text{Frac}(R_1)$. Thus R_1 is finitely generated over $\mathbb{Z}$ and K_1 is a finite extension of $\mathbb{Q}$, $\text{rank}_{\mathbb{Z}}(R_1) = [K_1 : \mathbb{Q}]$, and K_1 is the fraction field of R_1 .

Inductive Hypothesis: Assume that $\forall n \in \mathbb{N}$, R_{n-1} is finitely generated over $\mathbb{Z}$ and K_{n-1} is a finite extension of $\mathbb{Q}$, $\text{rank}_{\mathbb{Z}}(R_{n-1}) = [K_{n-1} : \mathbb{Q}]$, and K_{n-1} is the fraction field of R_{n-1} .

Inductive Step: Since a_n is a solution to F , it is algebraic over $\mathbb{Q}$ and thus algebraic over K_{n-1} . Thus $\exists f_n(X) \in K_{n-1}[X]$, an irreducible polynomial where a_n is a root. $f_n(x) = \sum_{k=0}^{\deg f_n} \frac{p_k}{q_k} x^k$ for $p_k, q_k \in R_{n-1} \ni q_k \neq 0$, since K_{n-1} is the fraction field of R_{n-1} . WLOG, $q_k = 1$, as you can multiply $f_n(x)$ by $\prod_{k=0}^{\deg f_n} q_k \in R_{n-1}$, and f_n would stay irreducible. Thus f_n irreducible in R_{n-1} . Thus $K_n \cong K_{n-1}[X]/(f_n(X))$ and $R_n \cong R_{n-1}[X]/(f_n(X))$. Since a_n is algebraic over $\mathbb{Q}$ and hence it is algebraic over K_{n-1} , K_{n-1} is a finite extension over $\mathbb{Q}$. By the inductive Hypothesis, since K_{n-1} is a finite extension over $\mathbb{Q}$, K_n is a finite extension over $\mathbb{Q}$. Similarly, R_n is finitely generated over R_{n-1} , and since $\mathbb{Z}[a_1, \dots, a_{n-1}]$ is finitely generated over $\mathbb{Z}$ (inductive hypothesis), R_n is finitely generated over $\mathbb{Z}$. Moreover $\{1, a_n, \dots, a_n^{\deg f_n}\}$ was a basis for R_n over R_{n-1} moreover since f_n was irreducible in K_{n-1} , that means, $\text{rank}_{R_{n-1}}(R_n) = \deg f_n$. $[K_n : K_{n-1}] = \deg f_n \implies [K_n : \mathbb{Q}] = \deg f_n * [K_{n-1} : \mathbb{Q}]$ and $\text{rank}_{\mathbb{Z}}(R_n) = \deg f_n * \text{rank}_{\mathbb{Z}}(R_{n-1})$. By the inductive hypothesis, $\text{rank}_{\mathbb{Z}}(R_{n-1}) = [K_{n-1} : \mathbb{Q}] \implies [K_n : \mathbb{Q}] = \text{rank}_{\mathbb{Z}}(R_n)$. $R_n \subset K_n$. Furthermore, $\forall k \in K_n$, $k = \frac{p(a_n)}{q(a_n)}$ for $p(x), q(x) \in R_{n-1}[x]$ since K_{n-1} is the fraction field of R_{n-1} and $q(a) \neq 0$, thus $k \in \text{Frac}(K_n)$. Thus $K_n \subseteq \text{Frac}(R_n)$. $\forall k \in \text{Frac}(R_n), k = \frac{r}{s}$ for $r, s \in R_n \ni s \neq 0, k \in K_n$ by definition. Thus $\text{Frac}(R_n) \subseteq K_n \implies K_n = \text{Frac}(R_n)$. Thus R_n is finitely generated over $\mathbb{Z}$ and K_n is a finite extension of $\mathbb{Q}$, $\text{rank}_{\mathbb{Z}}(R_n) = [K_n : \mathbb{Q}]$, and K_n is the fraction field of R_n .

Thus K is finitely generated over $\mathbb{Z}$, K is a finite extension over $\mathbb{Q}$, $\text{rank}_{\mathbb{Z}}(R) = [K : \mathbb{Q}]$, and K is the fraction field over R . $\square$

Problem 2. Explain why pR is not R . Explain why there is a maximal ideal P of R whose intersection with Z is pZ .

Proof. Let $G(x_1, \dots, x_k) \in \mathbb{Z}[x_1, \dots, x_k] \ni G(a_1, \dots, a_k) = 0$. Thus $R = \frac{\mathbb{Z}[x_1, \dots, x_k]}{(G(x_1, \dots, x_k))}$. To show why pR is not R it suffices to show that R/pR is not trivial.

$$\begin{aligned} R/pR &\cong \frac{\frac{\mathbb{Z}[x_1, \dots, x_k]}{(G(x_1, \dots, x_k))}}{p\left(\frac{\mathbb{Z}[x_1, \dots, x_k]}{(G(x_1, \dots, x_k))}\right)} \\ &\cong Z[x_1, \dots, x_k]/(p, G(x_1, \dots, x_k)) \\ &\cong Z[x_1, \dots, x_k]/p/(G(x_1, \dots, x_k)) \\ &\cong \mathbb{F}_p[x_1, \dots, x_k]/(\overline{G}(x_1, \dots, x_k)) \quad (\overline{G}(x_1, \dots, x_k) = G(x_1, \dots, x_k) \pmod{p}) \end{aligned}$$

Thus, $R/pR = \mathbb{F}_p[x]/(\overline{G}(x_1, \dots, x_k))$, which is not trivial. Thus $pR \neq R$.

Let $T :=$ set of proper ideals of R . $\exists B \in T \ni B = \bigcup_{I \in T} I$, every ideal in T is contained in B and thus B is an upperbound of T using containment as the ordering. By Zorn's Lemma, $\exists$ a maximal ideal $M \in T$, such that $\forall$ ideals I such that $M \subseteq I \subseteq R \implies I = M$ or $I = R$.

By Correspondence Theorem for Rings, the maximal ideals of $R/pR \cong \mathbb{F}_p[x_1, \dots, x_k]/(\overline{G}(x_1, \dots, x_k))$ correspond to the maximal ideals of R which contain pR (Note R/pR contains a maximal ideal using the same argument as above using Zorn's Lemma). Thus $\exists P$, a maximal ideal, where $pR \subseteq P \implies p\mathbb{Z} \subseteq P \implies p\mathbb{Z} \subseteq P \cap \mathbb{Z}$. $P \cap \mathbb{Z}$ is an ideal of $\mathbb{Z}$. By the maximality of $p\mathbb{Z}$, $p\mathbb{Z} = P \cap \mathbb{Z}$ or $P \cap \mathbb{Z} = \mathbb{Z}$. If $p\mathbb{Z} = P \cap \mathbb{Z}$, we are done. Assume the contrary $P \cap \mathbb{Z} = \mathbb{Z}$, thus $\mathbb{Z} \subseteq P$. $\forall r \in R$, $1 * r \in P \implies R \subseteq P \implies R = P$ which is a contradiction since P is proper.

Thus $P \cap \mathbb{Z} = p\mathbb{Z}$. □

Problem 3. Explain why $\text{Gal}(F)$ acts naturally on K , preserving the subring R . Let S be the set of maximal ideals of R whose intersection with Z is the prime ideal pZ . Explain why $\text{Gal}(F)$ acts naturally on S . Explain why for any J in S , the map $Z \rightarrow R/J$ has kernel (p) , so we get $\mathbb{F}_p \subset R/J$. Explain why R/J is a finite field, and why it is a splitting field for f .

Claim 3a. $\text{Gal}(F)$ acts naturally on K , preserving the subring R .

Proof. $\text{Gal}(F)$ acts as the following, $\forall g \in \text{Gal}(F)$, $\forall r \in K$, $g \cdot r = g(r)$.

$\forall g \in \text{Gal}(F)$, g fixes the elements of $\mathbb{Q}$ and permutes the roots of F , ie $a_1, \dots, a_k$. Since $\mathbb{Z} \in \mathbb{Q}$, g fixes the elements of $\mathbb{Z}$ and permutes the roots of F , $a_1, \dots, a_k$. Thus $\forall r \in R$, $r \in K$, and $r = \sum_{n=1}^d c_n \prod_{i=1}^k a_i^{n_i}$ for $c_n \in \mathbb{Z}$. Thus $g(r) = g(\sum_{n=1}^d c_n \prod_{i=1}^k a_i^{n_i}) = \sum_{n=1}^d c_n \prod_{i=1}^k g(a_i)^{n_i}$ since $g(a_i)$ is just another root of F , $g(r) \in R$. Thus R is preserved by the $\text{Gal}(F)$ action.

Furthermore, $\forall g \in \text{Gal}(F)$, $\forall r \in R$, $r = \sum_{n=1}^d c_n \prod_{i=1}^k a_i^{n_i}$ for $c_n \in \mathbb{Z}$. Let $s = \sum_{n=1}^d c_n \prod_{i=1}^k g^{-1}(a_i)^{n_i}$ and $g(s) = r$. Thus g is a ring automorphism of R . □

Claim 3b. Let S be the set of maximal ideals of R whose intersection with Z is the prime ideal pZ . $\text{Gal}(F)$ acts naturally on S .

Proof. $\forall g \in \text{Gal}(F)$, g is a automorphism of K which preserves R . $\forall J \in S$, let I be an ideal of R such that $g^*(J) \subset I \subset R$. That would mean, $J \subseteq g^{-1}(I) \subseteq R$. By the maximality of J , $g^{-1}(I) = J$ or $g^{-1}(I) = R$. If $g^{-1}(I) = J \implies I = g(J)$ and if $g^{-1}(I) = R \implies I = R$. Thus $g(J)$ is maximal. Since $g \in \text{Gal}(F)$, it is the identity on $\mathbb{Q}$ and since $\mathbb{Z} \subset \mathbb{Q} \implies g|_{\mathbb{Z}} = \text{id}$. This means $g(p\mathbb{Z}) = p\mathbb{Z}$, more specifically it fixes each element. Since $g(p\mathbb{Z}) \subset g(J) \implies p\mathbb{Z} \subset g(J)$. Thus $g(J) \in S$ by definition.

Define the action of $\text{Gal}(F)$ on S as the following, $\forall J \in S$, $g \cdot J = g_*(J) = g(J)$. □

Problem 3c. Explain why for any J in S , the map $Z \rightarrow R/J$ has kernel (p) , so we get $F_p \subset R/J$.

Proof. $\forall J \in S$, The map $\varphi : \mathbb{Z} \rightarrow R/J$ is the restriction of the natural quotient map, $\pi : R \rightarrow R/J$ to $\mathbb{Z}$. The $\ker \pi = J$. Thus $\ker \varphi = \ker \pi \cap \mathbb{Z} = J \cap \mathbb{Z} = p\mathbb{Z}$, by definition of J . Thus the map φ factors through the map $\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$, and since all the kernel is killed of by φ in this map, $\mathbb{F}_p \subset R/J$. $\square$

Problem 3d. Explain why R/J is a finite field, and why it is a splitting field for f .

Proof. $\forall r \in R, \bar{r} := r + J$. Since $a_1, \dots, a_k$ are the generators of R . Then $\bar{a}_1, \dots, \bar{a}_k$ are generators of R/J .

$\forall r \in R$, define $\bar{r} = r + J$.

Take $q : R \rightarrow R/J$, where $r \rightarrow \bar{r}$ the standard quotient map. Since R/J is a finite extension of $\mathbb{F}_p$, $\mathbb{F}_p \leq R/J$, where $\forall z \in \mathbb{Z}, [z] \rightarrow \bar{z}$. Thus $f([x]) = [F([x])] = \overline{F(\bar{x})}$. Suppose, $F(x) = \sum_{i=0}^{\deg F} c_n x^n$

Thus $\forall i \in [1, \dots, k], f(\bar{a}_i) = \overline{\sum_{i=0}^{\deg F} c_n \bar{a}_i^n} = \overline{\sum_{i=0}^{\deg F} c_n a_i^n} = \bar{0}$. Thus a_i is a root of f .

Thus there is a map p from roots of F to roots of f where $a_i \rightarrow \bar{a}_i$. Since q is a ring homomorphism and $F(x) = \prod_{i=1}^k (x - a_i) \implies q(F(x)) = f(x) = \prod_{i=1}^k (x - \bar{a}_i)$. Assume the contrary p is not injective, then $\exists i, j \in [1, \dots, k] \ni \bar{a}_j = \bar{a}_i$, thus $(x - \bar{a}_i)^2 | f(x) \implies f(x)$ has a double root a_i which is a contradiction. Thus p is injective.

Let r be a root of f , then $(x - r) | f \implies (x - r) | \prod_{i=1}^k (x - \bar{a}_i) \implies \exists i \in [1, \dots, k] \ni x - r = x - \bar{a}_i \implies r = \bar{a}_i \implies a_i = p(r)$. Thus p is surjective.

Thus $p_*(\{a_1, \dots, a_k\}) = \{\bar{a}_1, \dots, \bar{a}_k\}$ are all the roots of f and they are distinct. Thus $\bar{a}_1, \dots, \bar{a}_k$ are algebraic over $\mathbb{F}_p$. Thus R/J is a finite extension over $\mathbb{F}_p$.

Since $\forall i \in [1, \dots, k], \bar{a}_i \in R/J \implies f$ splits over R/J . Moreover since $\bar{a}_1, \dots, \bar{a}_j$ are generators of R/J thus R/J is the splitting field of f . $\square$

Problem 4. Let $P_1, \dots, P_k$ be any finite collection of distinct elements from S . Explain why the natural map $\varphi : R \rightarrow R/P_1 \times R/P_2 \times \dots \times R/P_k$ is surjective (feel free to look up the Chinese remainder theorem) and why this factors through R/pR .

Proof. $\forall I, J \leq R$ where I and J are ideals of R . $\forall r \in I + J, r = i + j$ for some $i \in I$ and $j \in J$. Thus $\forall i_1 + j_1, i_2 + j_2 \in I + J$ where $i_1, i_2 \in I$ and $j_1, j_2 \in J, i_1 + j_2 + i_2 + j_2 = (i_1 + i_2) + (j_1 + j_2) \in I + J$. $\forall r \in R, \forall i + j \in I + J, r(i + j) = r(i) + r(j), r(i) \in I$ and $r(j) \in J$ by definition of ideals. Thus $r(i + j) \in I + J$. Thus $I + J$ is an ideal.

$\forall i, j \in [1, \dots, k] \ni i \neq j, P_i$ and P_j are distinct. $P_i \subseteq P_i + P_j \subseteq R$ and $P_j \subseteq P_i + P_j \subseteq R$. By the maximality of $P_i, P_i + P_j = R$ or $P_i + P_j = P_i$. By the maximality of $P_j, P_i + P_j = R$ or $P_i + P_j = P_j$. Assume the contrary that $P_i + P_j = P_i$, then $P_i + P_j \neq R$ but since $P_i \neq P_j$ (since they are distinct), then $P_i + P_j \neq P_j$ which contradicts the maximality of P_j . Thus $P_i + P_j \neq P_i$. Thus $P_i + P_j = R$. Thus by Chinese Remainder Theorem, $\varphi : R \rightarrow R/P_1 \times \dots \times R/P_k$ is surjective.

$\ker \varphi$, is by definition the elements of R which map to the identity on R/P_i for all i . In other words, its the elements of R which are part of each P_i , or the intersection of all the P_i . Thus $\ker \varphi = \bigcap_{i=1}^k P_i$. By definition of $S, \forall i \in [1, \dots, k], p\mathbb{Z} \subset P_i \implies p \in P_i \implies pR \subset P_i$, by definition of ideals. Thus $pR \subset \bigcap_{i=1}^k P_i = \ker \varphi$. Thus $\ker[R \rightarrow R/pR] \subset \ker \varphi$. By prehomework, φ factors through $R \rightarrow R/pR$. $\square$

Problem 5. Explain why R/pR is a finite dimensional $\mathbb{F}_p$ vector space of dimension d (recall is the rank of R over Z and also the $\dim$ of K over Q)

Claim 5a. $R/pR \cong (\mathbb{Z}/(p))^d \cong F_p^d$

Proof. Note elements of $\mathbb{F}_p$ will be denoted as $[z]$ (equivalence classes) for $z \in \mathbb{Z}$.

By problem 1, $R \cong \mathbb{Z}^d$ as a $\mathbb{Z}$ module. Thus $\forall r \in R, r = (r_1, \dots, r_n)$ for $r_1, \dots, r_n \in \mathbb{Z}$. Define the map, $\varphi : R/pR \rightarrow (\mathbb{Z}/(p))^d$ where $(r_1, \dots, r_n) + pR \rightarrow ([r_1], \dots, [r_n])$.

$\forall (r_1, \dots, r_n) + pR \in R/pR, \forall (r_1 + ps_1, \dots, r_1 + ps_1), (r_1 + pt_1, \dots, r_1 + pt_1) \in (r_1, \dots, r_n) + pR$, $\varphi((r_1 + ps_1, \dots, r_n + ps_n) + pR) = ([r_1 + ps_1], \dots, [r_n + ps_n]) = ([r_1], \dots, [r_n]) = ([r_1 + pt_1], \dots, [r_n + pt_n]) = \varphi((r_1 + pt_1, \dots, r_n + pt_n) + pR)$. Thus φ is well defined.

$\forall \alpha, \beta \in \mathbb{Z}$, and $(r_1, \dots, r_n) + pR, (s_1, \dots, s_n) + pR \in R/pR$, $\varphi(\alpha((r_1, \dots, r_n) + pR) + \beta((s_1, \dots, s_n) + pR)) = \varphi((\alpha r_1 + \beta s_1, \dots, \alpha r_n + \beta s_n) + pR) = ([\alpha r_1 + \beta s_1], \dots, [\alpha r_n + \beta s_n]) = [\alpha]([r_1], \dots, [r_n]) + [\beta]([s_1], \dots, [s_n]) = [\alpha]\varphi((r_1, \dots, r_n) + pR) + [\beta]\varphi((s_1, \dots, s_n) + pR)$. Thus φ is a homomorphism.

$\forall k \in \ker \varphi, k = (k_1, \dots, k_n) + pR$ where $\varphi(k) = ([0], \dots, [0]) \implies k_i \mod p \equiv 0$. Thus $k = pR$. φ is an injective map.

$\forall ([r_1], \dots, [r_n]) \in (\mathbb{Z}/(p))^d$, take $s_i \in [r_i]$ for all i . $s_i \in \mathbb{Z}$. $(s_1, \dots, s_n) + pR \in R/pR$ and $\varphi((s_1, \dots, s_n) + pR) = ([r_1], \dots, [r_n]) \in (\mathbb{Z}/(p))^d$. Thus the map is surjective. Thus the map is an isomorphism.

Thus $R/pR \cong (\mathbb{Z}/(p))^d$. By definition, $\mathbb{F}_p \cong \mathbb{Z}/(p)$. Thus $R/pR \cong \mathbb{F}_p^d$. Thus R/pR is a $\mathbb{F}_p$ vector space with dimension d . $\square$

Problem 6. Explain why the rings R/J , for $J \in G \cdot P$ are all isomorphic finite fields, containing F_p . Say $|G \cdot P| = m$ and let k be the degree of each R/J over F_p . Explain why $d \geq mk$

Proof. $\forall J \in G \cdot P, \forall g \in G$, define the maps $\alpha : R/J \rightarrow R/g(J)$ where $r + J \rightarrow g(r) + g(J)$ and $\beta : R/g(J) \rightarrow R/J$ where $r + g(J) \rightarrow g^{-1}(r) + J$. The maps are valid because g is an automorphism of R . $\forall r + J \in R/J, \beta(\alpha(r + J)) = \beta(g(r) + g(J)) = g^{-1}(g(r)) + J = r + J$. Thus $\beta \circ \alpha = id|_{R/J}$. $\forall r + g(J) \in R/g(J), \alpha(\beta(r + g(J))) = \alpha(r + J) = r + g(J)$ thus $\alpha \circ \beta = id|_{R/g(J)}$. $\forall r + J, s + J \in R/J$ and $\alpha(r + J + s + J) = \alpha((r + s) + J) = g(r + s) + J = g(r) + g(s) + J = g(r) + J + g(s) + J = \alpha(r + J) + \alpha(s + J)$. $\alpha((r + J)(s + J)) = \alpha(rs + J) = g(r)g(s) + J = g(r)g(s) + g(r)g(J) + g(s)g(J) + g(J) = (g(r) + g(J))(g(s) + g(J)) = \alpha(r + J)\alpha(s + J)$. Thus $R/J \cong R/g(J)$.

Furthermore, $\forall g \in G, g(J)$ is a maximal ideal of R . Thus $R/g(J)$ is a field. By 3c, $F_p \subseteq R/J$.

Let $G \cdot P = \{P_1, \dots, P_k\}$. Let $\varphi : R \twoheadrightarrow R/P_1 \times \dots \times R/P_k$. By problem 4, φ is surjective. Moreover φ factors through $\rho : R \twoheadrightarrow R/pR$. Thus there is a map, $\sigma : R/pR \twoheadrightarrow R/P_1 \times \dots \times R/P_k$. Since R/P_1 is a finite extension of degree k over F_p it is also a vector space over F_p with dimension k . Thus $R/P_1 \times \dots \times R/P_k$ is a vectorspace with dimension, mk . By rank nullity theorem, $\dim R/pR = \dim \sigma_*(R/pR) + \dim \ker \sigma = mk + \dim \ker \sigma \implies d = \dim R/pR \geq mk$. Thus $d \geq mk$. $\square$

Problem 7. Explain why the natural quotient map $R \rightarrow R/J$ induces a bijection between the roots of F and the roots of f , and why R/J is a splitting field of f .

Proof. $\forall r \in R$, define $\bar{r} = r + J$.

Take $q : R \rightarrow R/J$, where $r \rightarrow \bar{r}$ the standard quotient map. Since R/J is a finite extension of $\mathbb{F}_p$, $\mathbb{F}_p \leq R/J$, where $\forall z \in \mathbb{Z}, [z] \rightarrow \bar{z}$. Thus $f([x]) = [F([x])] = \overline{F(\bar{x})}$. Suppose, $F(x) = \sum_{i=0}^{\deg F} c_n x^n$

Thus $\forall i \in [1, \dots, k], f(\bar{a}_i) = \sum_{i=0}^{\deg F} c_n \bar{a}_i^n = \sum_{i=0}^{\deg F} c_n a_i^n = \bar{0}$. Thus a_i is a root of f .

Thus there is a map p from roots of F to roots of f where $a_i \rightarrow \bar{a}_i$. Since q is a ring homomorphism and $F(x) = \prod_{i=1}^k (x - a_i) \implies q(F(x)) = f(x) = \prod_{i=1}^k (x - \bar{a}_i)$. Assume the contrary p is not injective,

then $\exists i, j \in [1, \dots, k] \ni \overline{a_j} = \overline{a_i}$, thus $(x - \overline{a_i})^2 | f(x) \implies f(x)$ has a double root a_i which is a contradiction. Thus p is injective.

Let r be a root of f , then $(x - r) | f \implies (x - r) | \prod_{i=1}^k (x - \overline{a_i}) \implies \exists i \in [1, \dots, k] \ni x - r = x - \overline{a_i} \implies r = \overline{a_i} \implies a_i = p(r)$. Thus p is surjective.

Thus $p_*(\{a_1, \dots, a_k\}) = \{\overline{a_1}, \dots, \overline{a_k}\}$ are all the roots of f and they are distinct.

Since $\forall i \in [1, \dots, k], \overline{a_i} \in R/J \implies f$ splits over R/J . Moreover since $\overline{a_1}, \dots, \overline{a_j}$ are generators of R/J thus R/J is the splitting field of f . $\square$

Problem 8. Let G^P be the stabilizer of G (for the natural action of G on S described in 3). Explain why there is a natural homomorphism $G^P \rightarrow \text{Gal}(f)$, and why this is injective. Explain why $|G^P| \leq k$ (k is defined in 6))

Proof. $\forall g \in G$, define $\overline{g} : R/P \rightarrow R/g(P)$ where $r + P \mapsto r + g(P)$. $\overline{g}$ is the map α in problem 6 and thus is a isomorphism between, R/P and $R/g(P)$.

$\forall g \in G^P$, $\overline{g}$ is a isomorphism of G/P and $G/g(P) = G/P$ and thus a automorphism of G/P . Furthermore, $\forall r \in [0, \dots, p-1], \overline{g}(r+P) = g(r)+P, g(r) = r$, since $g \in \text{Gal}(F)$. Thus $\overline{g}(r+P) = r+P$. Thus $\overline{g}$ fixes elements of $\overline{F_p} \in R/P$ and thus $\overline{g} \in \text{Gal}(f)$. Let $\varphi : G^P \rightarrow \text{Gal}(f)$, where $g \mapsto \overline{g}$. $\forall g, h \in G^P, \varphi(g \circ h) = \overline{g \circ h} = \overline{g} \circ \overline{h}$. $\forall r+P \in R/P, \overline{g \circ h}(r+P) = g(h(r))+P = \overline{g}(h(r)+P) = (\overline{g} \circ \overline{h})(r+P)$. Thus $\varphi(g \circ h) = \overline{g \circ h} = \overline{g} \circ \overline{h} = \varphi(g) \circ \varphi(h)$. Thus φ is a homomorphism.

$\forall g \in \ker \varphi$, any element of G is determined by its permutation it induces on the roots of F , $a_1, \dots, a_k$. Thus, $\forall i \in [1, \dots, k], \varphi(g)(a_i + P) = \overline{g(a_i)} + P = \overline{g(a_i)} = a_i$. Assume the contrary that $g \neq id$, then $\exists j \in [1, \dots, k] \ni g(a_j) \neq a_j \implies \overline{g(a_j)} \neq \overline{a_j}$ by 5 (bijection between roots of f and roots of F). Thus $g \notin \ker \varphi$ and a contradiction occurs. Thus g must be id . Thus φ is injective.

Since R/P is the splitting field of f , by the extension homework, $|\text{Gal}(f)| \leq [R/P : \mathbb{F}_p] = k$. Since φ is injective, $|G^P| \leq |\text{Gal}(f)| \leq k \implies |G^P| \leq k$. $\square$

Problem 9. Explain why $|G| = d$, and why $d = |G^P| |G^P|$.

Proof. Assume the contrary that F has double roots, wlog let a_1 be the root with multiplicity 1. Thus $F(a_1) = F'(a_1) = 0$. Then $f(\overline{a_1}) = f'(\overline{a_1}) = 0 \implies f$ has a double root over some extension. However this contradicts the assumption about f . Thus F has no double roots and $\forall i, j \in [1, \dots, k], i \neq j \implies a_i \neq a_j$. Thus F is seperable over $\mathbb{Q}$.

There are 2 cases now:

1. F is irreducible: If F is irreducible, it is the irreducible polynomial of $a_1, \dots, a_k$ and splits completely over K and is seperable thus K is galois (irreducibles of generators splitting and seperable $\implies$ irreducibles of all elements split and are seperable over K , proven in a previous homework).
2. F is not irreducible: If F is not irreducible, then $\exists g_1, \dots, g_n \in \mathbb{Q}[X]$ irreducible where $F(x) = g_1(x) \dots g_n(x)$. Assume the contrary that $\exists i, j \in [1, \dots, n] \ni g_i$ and g_j are not coprime and share a factor. Thus $\exists a_i \ni g_i(a_i) = 0$ and $g_j(a_i) = 0$. Thus $(x - a_i) | g_i$ and $(x - a_i) | g_j$ thus $(x - a_i) | F(x)$. Thus $F(x)$ has a repeated root which is a contradiction because F is seperable. Thus $g_1, \dots, g_n$ are coprime and thus $\forall i \in [1, \dots, k], a_i$ is a root of 1 and only 1 $g_j(x)$. Since F is seperable, g_j is seperable, $\forall j \in [1, \dots, n]$. Since F splits over K , g_j splits over K . Thus, $\forall i \in [1, \dots, k]$, the irreducible polynomial of a_i splits completely over K and is seperable. Thus K is galois.

Since K is galois, $|G| = [K : \mathbb{Q}] = d$. By Orbit Stabilizer Theorem, $d = |G| = |G^P| |G \cdot P| = m|G^P|$. $\square$

Problem 10. Using the above, show that $G^P \rightarrow \text{Gal}(f)$ must be an isomorphism.

Proof. By problem 9, $d = |G^P|m \implies |G^P| = \frac{d}{m}$. By problem 6, $d \geq mk \implies k \leq d/m = |G^P|$. By problem 8, $|G^P| \leq k$. Thus $|G^P| = k$. However, $|\text{Gal}(f)| \leq k$. But $k = |G^P| \leq |\text{Gal}(f)| \leq k \implies |G^P| = |\text{Gal}(f)| = k$. Since $G^P \rightarrow \text{Gal}(f)$ is an inclusion and since $|G^P| = |\text{Gal}(f)|$, the map is also surjective thus an isomorphism exists between, $G^P \rightarrow \text{Gal}(f)$. Thus $G^P \cong \text{Gal}(f)$ $\square$

Problem 11. Explain why G acts transitively on S , ie why $S = G \cdot P$ (ie there is just one orbit for the action of G on S).

Proof. Assume the contrary that G doesn't act transitively on S . Thus $\exists Q \in S \ni \forall g \in G, g \cdot P \neq Q$. More generally that $\forall g, h \in G, g(P) \neq h(Q)$. As proven in problem 4, $R \rightarrow R/g(P) \times R/g(Q)$ is surjective for all $g \in G$, since $g(P) \in S$ and $g(Q) \in S$. Thus $\exists r \in R \ni r \rightarrow (0 + g(P), 1 + g(Q))$. Thus $r \in g(P)$ for all g and $r \notin g(Q)$ for all g . Let $G_r = \prod_{g \in G} g(r)$. Assume the contrary that $\exists g \in G \ni g(r) \in Q$, then $r \in g^{-1}(Q)$ which is a contradiction. Thus $\forall g \in G, g(r) \notin Q$, moreover since $r \rightarrow 1 + g(Q), \forall h \in G, h(r) \rightarrow h(1) + g(Q) = 1 + g(Q)$. Since $R \rightarrow R/g(Q)$ is a homomorphism of rings, $G_r \rightarrow (1 + g(Q)) \dots (1 + g(Q)) = 1 + g(Q) \implies G_r \notin g(Q) \implies G_r \notin Q. \forall h \in G, h(G_r) = h(\prod_{g \in G} g(r))$ which simply reorders the product thus $h(G_r) = G_r$. Thus G_r is fixed by G thus $G_r \in \mathbb{Q}$. But since G preserves R and $r \in R$ and R is closed under multiplication, thus $G_r \in R$. Thus $G_r \in R \cap \mathbb{Q} = \mathbb{Z}$. Since $r \in \text{id}(P) \implies r \in P$, thus by definition of ideals $G_r \in P \implies G_r \in P \cap \mathbb{Z} = p\mathbb{Z}$. However $p\mathbb{Z} \subset Q$, which contradicts the fact that $G_r \notin Q$. Thus by contradiction, G must act transitively on S . $\square$

Problem 12. Explain why there is a copy of $\text{Gal}(f)$ inside $\text{Gal}(F)$. Is it necessarily unique? If not, What is it unique up to.

Proof. Since $\forall P \in S, G^P \cong \text{Gal}(f)$ and since $G^P \leq \text{Gal}(F) \implies \text{Gal}(f) \leq \text{Gal}(F)$ through G^P . However since $S \cong G \cdot P$, by our favorite formula (stabilizer of gP is conjugate to stabilizer of P), $\forall T \in G \cdot P, \exists g \in G \ni T = g(P)$ and $G^T = G^{g \cdot P} = gG^P g^{-1}$. Thus $\text{Gal}(f) \leq \text{Gal}(F)$ is unique up to conjugation. $\square$

Problem 13. Why is this cool?

Proof. This is cool as it allows us to learn more about the galois group of the splitting field of a polynomial $F \in \mathbb{Z}[x]$. By studying the galois group of the the splitting field of the $F \bmod p$ over a finite field, we can understand the subgroup structure of the galois group since $\text{Gal}(F \bmod p) \leq \text{Gal}(F)$. Moreover the fact that all copies of $\text{Gal}(F \bmod p)$ are conjugate, gives us some additional conjugacy structure which we can exploit to study the group. This fact paired with the Sylow Theorems may allow us to figure out what a group is. Moreover $\text{Gal}(F \bmod p)$ is likely much simpler due to the additional structure provided by finite fields. Overall the structure provided by the $\text{Gal}(f)$ s would help us find and identify the galois group of the splitting field of F or rule out possible groups which they cannot be, using the fact that different copies of $\text{Gal}(f)$ are conjugate. $\square$